

Joshua Shinkle

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EDUCATION

Purdue University, Ph.D. in Computer Science 2024 - 2029 (expected)

- Advisor: Associate Professor Antonio Bianchi

Taylor University, B.S. in Computer Science - Cybersecurity (*summa cum laude*) 2020 - 2024

- Advisor: Dannie Stanley, Ph.D.

ACADEMIC APPOINTMENTS

Purdue University - Graduate Research Assistant - PurSec Lab 2024 - Present

- Developed a contention-based timing side-channel for mobile NPUs
- Investigated the privacy risks of military-marketed mobile apps
- Produced Android userspace exploits for DARPA's Intelligent Generation of Tools for Security (INGOTS) program

RESEARCH INTERESTS

My research interests are in software/systems security and privacy. My research leverages program analysis, reverse engineering, and human-centered methods to study vulnerabilities in mobile phones across both the hardware and software layers.

PEER-REVIEWED CONFERENCE PUBLICATIONS

Joshua Shinkle, Chandrika Mukherjee, Abdullah Imran, Arjun Arunasalam, Donna Artusy, Antonio Bianchi, Z. Berkay Celik, Alexander Master

No Privacy for Privates: How Military Communities Experience and Perceive the Privacy Risks of Military-Marketed Mobile Apps

In Proceedings on Privacy Enhancing Technologies (PoPETs), pages 772–792, 2026

The Andreas Pfitzmann Best Student Paper Award and Best Presentation Award

AWARDS AND HONORS

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- 2026, Best Presentation Award for the paper “No Privacy for Privates: How Military Communities Experience and Perceive the Privacy Risks of Military-Marketed Apps” at Privacy Enhancing Technologies Symposium (PETS).
 - 2026, The Andreas Pfitzmann Best Student Paper Award for the paper “No Privacy for Privates: How Military Communities Experience and Perceive the Privacy Risks of Military-Marketed Apps” in Proceedings on Privacy Enhancing Technologies (PoPETs).

PRESENTATIONS

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- No Privacy for Privates: How Military Communities Experience and Perceive the Privacy Risks of Military-Marketed Apps, Privacy Enhancing Technologies Symposium (PETS). Calgary, AB, Canada, July 2026
 - Smartphone OPSEC Risk: Mobile Apps Targeting U.S. Service Members and Their Families, AvengerCon X, MD, USA, June 2026

INDUSTRY EXPERIENCE

Peraton Labs - Cybersecurity Research Internship Summer '26

- Implemented novel sensor fusion capabilities on Android phones by leveraging Bayesian inference networks

Peraton Labs - Cybersecurity Research Internship*Summer '25*

- Built an Android utility that leverages kernel tracing and interprocess communication introspection to trace messages between apps, telephony, Radio Interface Layer (RIL), and baseband firmware

Peraton Labs - Cybersecurity Research Internship*Summer '24*

- Trained a multitask deep learning recommender system to predict mobile app usage based on user demographics
- Worked as part of a team to repurpose mobile phone communication channels to enable secure communications

Peraton Labs - Cybersecurity Research Internship*Summer '23*

- Built a python tool to schedule/generate Android automations to mimic human activity
- Worked as part of a team to repurpose mobile phone communication channels to enable secure communications

Lockheed Martin ATL - Cybersecurity Research Internship*Summer '22*

- Reverse engineered Windows applications
- Windows kernel debugging

Anchorage Digital - Software Engineering Internship*Summer '21*

- Worked as part of a team to prototype a new feature critical to supporting DeFi apps on the Ethereum blockchain